

Digital Logic Final Report

1. Truth Table 1

Table inputs:

There are 2 inputs A0 & B0 and two selectors

X & Y which have these combinations:

(0, 0)

(0, 1)

(1, 0)

(1, 1)

x	y	A0	B0		R0
0	0	0	0	0 AND	0
0	0	0	0	1 AND	0
0	0	1	0	0 AND	0
0	0	1	1	1 AND	1
0	1	0	0	0 OR	0
0	1	0	1	1 OR	1
0	1	1	1	0 OR	1
0	1	1	1	1 OR	1
1	0	0	0	0 XOR	0
1	0	0	1	1 XOR	1
1	0	1	0	0 XOR	1
1	0	1	1	1 XOR	0
1	1	0	0	0 XNOR	1
1	1	0	1	1 XNOR	0
1	1	1	0	0 XNOR	0
1	1	1	1	1 XNOR	1

When X=0,Y=0 are selected the AND gate will be used between A0 and B0

When X=0,Y=1 are selected the OR gate will be used between A0 and B0

When X=1,Y=0 are selected the XOR gate will be used between A0,B0

When X=1,Y=1 are selected the XNOR gate will be used between A0,B0

Logic gates and their functions:

2. Truth Table 2

We got here 2 inputs A1 & B1 , 2 selector X & Y and 1 output which is R1.

As we can illustrate we did 2 tables (8-bits) so it could be easier and it helped us to decrease the number of k-maps used.

x	y	A1	B1		R1
0	0	0	0	0 AND	0
0	0	0	0	1 AND	0
0	0	1	0	0 AND	0
0	0	1	1	1 AND	1
0	1	0	0	0 OR	0
0	1	0	1	1 OR	1
0	1	1	1	0 OR	1
0	1	1	1	1 OR	1
1	0	0	0	0 XOR	0
1	0	0	1	1 XOR	1
1	0	1	1	1 XOR	0
1	0	1	0	0 XOR	1
1	1	0	0	1 XNOR	0
1	1	0	1	0 XNOR	0
1	1	1	1	1 XNOR	1

Conclusion:

These tables describes the behavior of the logic units as we listed all the possible combinations of A0,B0 and A1,B1 and their selectors X,Y and their outputs R0,R1.

3. Karnaugh Maps (K-Maps)

. K-Map Structure

Variables:

Horizontal: X and Y (combined as XY)

Vertical: A1 and B1 (combined as A1B1)

K-Map Size:

4 columns: XY={00,01,11,10}

4 rows: A1B1={00,01,11,10}

XY \ A ₁ B ₁	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	1	1	0
10	0	1	0	1

Entries:

Each cell contains a binary output (0 or 1).

The values are results of a logic function based on the combinations of XY and A1B1

Grouping:

Several 1's are grouped in adjacent cells, allowing for minimization.

Functions:

$$F = XYA_1'B_1' + XY'A_1'B_1 + XY'A_1B_1' + YA_1B_1 + X'A_1B_1 + A_1X'Y'B_1X'Y$$

As we simplify this equation we get the following:

$$F = A_1B_1(Y+X') + X'Y(A_1+B_1) + XYA_1'B_1' + XY'(A_1 \oplus B_1)$$

Conclusion:

These k-maps makes it easy to simplify logic functions, helps reduce the number of logic gates in our circuit, shows patterns in the logic clearly and reduces errors compared to solving with Boolean algebra.

XY \ A ₁ B ₁	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	1	1	0
10	0	1	0	1

4. Logic Circuit

Circuit Diagram Explanation

Overview

The circuit diagram illustrates a logic unit designed to operate on two 2-bit binary numbers, A1 A0 and B1 B0. It performs four logic operations — AND, OR, XOR, and XNOR — using combinational logic gates and outputs the result based on the selected operation using two selectors x and y.

Inputs

The inputs are:

- A0 and B0: Operated on in the upper half of the circuit
- A1 and B1: Operated on in the lower half of the circuit

Logic Gates Used

Each pair of inputs is processed using the following gates:

- AND gate
- OR gate
- XOR gate
- XNOR gate

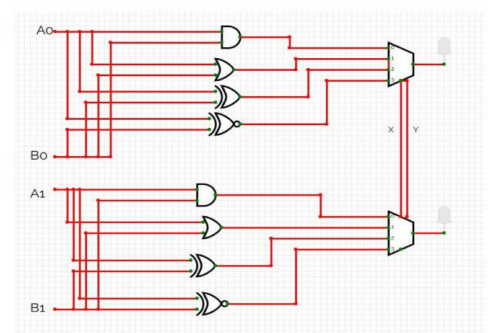
Multiplexers

Each half of the circuit contains a 4-to-1 multiplexer. The purpose of each multiplexer is to select one of the four logic operation outputs based on the values of selectors x and y. The selected output becomes the result for that specific bit pair (A0 B0 or A1 B1).

Control Logic

The two control selectors (x and y) determine which logic operation is selected by the multiplexer. The control logic follows this scheme:

- x = 0, y = 0: AND operation
- x = 0, y = 1: OR operation



- $x = 1, y = 0$: XOR operation
- $x = 1, y = 1$: XNOR operation

Output

The outputs from the two multiplexers are labeled as X and Y:

- r0 corresponds to the logic result of A0 and B0
- r1 corresponds to the logic result of A1 and B1

These two outputs together form the final 2-bit result of the logic unit.

Conclusion :

The diagram successfully implements a 2-bit logic unit using basic logic gates and multiplexers. By changing the values of the selectors x and y, the user can dynamically choose which logic operation is performed on the input bits.

5. Circuit

To design, implement, and test a digital logic circuit using **AND, OR, XOR, XNOR gates and a 4:1 multiplexer (MUX) with:

- Four input variables (`A0, B0, A1, B1`)
- Two selector lines (`X, Y`) for MUX control
- 5V adapter power supply
- LED output visualization with current-limiting resistors

2. Components & Setup:

2.1 Components List:

Component	Specifications	Quantity
AND Gate IC	7408 (Quad 2-input AND)	1
OR Gate IC	7432 (Quad 2-input OR)	1
XOR Gate IC	7486 (Quad 2-input XOR)	1
NOT Gate IC	7414 (Hex Inverter)	1
MUX IC	74153 (Dual 4:1 MUX)	1
LEDs	Red/Green (5mm)	2
Resistors	5 kΩ (for LEDs)	2
Breadboard	–	1
Toggle Switches	SPDT (for inputs)	4
5V DC Adapter	2.1mm barrel jack, 1A	1
Jumper Wires	Color-coded 	As needed

2.2 Wiring Scheme:

Signal	Wire Color	Connection
A0	Yellow	All gate inputs (AND/OR/XOR)
B0	Orange	All gate inputs (AND/OR/XOR)
A1	Brown	All gate inputs (AND/OR/XOR)
B1	Blue	All gate inputs (AND/OR/XOR)
X (S0)	Purple	MUX Selector Pin 1
Y (S1)	White	MUX Selector Pin 2
Vcc (5V)	Red	All ICs (Pin 14/16) + Power Rail
Ground	Black	All ICs (Pin 7/8) + Ground Rail
Gate Out	Green	MUX Inputs (I0–I3)

3. Circuit Implementation:

3.1 Gate Configurations:

Gate	IC	Input Pins	Output
AND	7408	1 (A0), 2 (B0)	Pin 3: A0 AND B0
		4 (A1), 5 (B1)	Pin 6: A1 AND B1
OR	7432	1 (A0), 2 (B0)	Pin 3: A0 OR B0
		4 (A1), 5 (B1)	Pin 6: A1 OR B1
XOR	7486	1 (A0), 2 (B0)	Pin 3: A0 XOR B0
XNOR	7486 + 7414	XOR Out → NOT In	Pin 2 (7414): XNOR

3.2 MUX (74153) Setup:

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MUX Input	Connected To
I0	AND Output
I1	OR Output
I2	XOR Output
I3	XNOR Output

- Selection Logic:

Y (S1)	X (S0)	Selected Output
0	0	AND
0	1	OR
1	0	XOR
1	1	XNOR

3.3 Power Supply (Adapter):

- Adapter Output: 5V DC → Breadboard power rail.
- Ground: Common ground rail for all ICs and LEDs.
- Safety:
- Fused (500mA) for overcurrent protection.
- Polarity verified with multimeter.

4. Testing & Result:

4.1 Test Cases:

A0	B0	A1	B1	Y	X	Selected Gate	LED State
0	0	1	1	0	0	AND (0)	OFF
1	0	1	1	0	1	OR (1)	ON
1	1	0	0	1	0	XOR (0)	OFF
1	0	0	1	1	1	XNOR (0)	OFF

4.2 Observations:

- LEDs lit only when selected MUX output = `1`.
- All gates responded correctly to input combinations.
- No voltage drops observed with adapter power.

5. Conclusion

- Successfully implemented a (4-input logic circuit) with MUX-based output selection.
- (5V adapter) provided stable power without voltage fluctuations.
- Color-coded wiring simplified debugging.
- Future work: Expand to 8:1 MUX or add a 7-segment display.

6. Appendix:

- Datasheets: [7408](https://www.ti.com/lit/ds/symlink/sn7408.pdf), [74153](https://www.ti.com/lit/ds/symlink/sn74153.pdf)

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